

05 - zz Routing Basics - EXERCISE (my solution)

Lesson 3 Exercises: Routing Basics & Static Routes

Complete these exercises to build your routing and troubleshooting skills.

Exercise 1: Deploy, Configure, and Verify End-to-End

Objective: Deploy the hub-and-spoke topology, apply Ansible configuration, and verify full connectivity.

Steps

1. Deploy the topology:

```
cd lessons/clab/03-routing-basics  
containerlab deploy -t topology/lab.clab.yml
```

✓ done

2. Wait for SR Linux to initialize (about 30 seconds), then apply config:

```
cd ansible  
ansible-playbook -i inventory.yml playbook.yml
```

✓ done

3. Verify host-to-router connectivity (same subnet):

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.1.1  
docker exec clab-routing-basics-host2 ping -c 3 10.1.4.1  
docker exec clab-routing-basics-host3 ping -c 3 10.1.5.1
```

4. Verify end-to-end cross-subnet connectivity (this is the lesson 02 cliffhanger payoff):

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.4.2  
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2  
docker exec clab-routing-basics-host2 ping -c 3 10.1.5.2
```

```
[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host1 ping -c 3 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=64 time=13.7 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=64 time=1.59 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=64 time=0.793 ms

--- 10.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.793/5.351/13.675/5.894 ms

[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host2 ping -c 3 10.1.4.1
PING 10.1.4.1 (10.1.4.1) 56(84) bytes of data.
64 bytes from 10.1.4.1: icmp_seq=1 ttl=64 time=1.49 ms
64 bytes from 10.1.4.1: icmp_seq=2 ttl=64 time=1.19 ms
64 bytes from 10.1.4.1: icmp_seq=3 ttl=64 time=0.870 ms

--- 10.1.4.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.870/1.184/1.489/0.252 ms

[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host3 ping -c 3 10.1.5.1
PING 10.1.5.1 (10.1.5.1) 56(84) bytes of data.
64 bytes from 10.1.5.1: icmp_seq=1 ttl=64 time=2.03 ms
64 bytes from 10.1.5.1: icmp_seq=2 ttl=64 time=1.84 ms
64 bytes from 10.1.5.1: icmp_seq=3 ttl=64 time=1.55 ms

--- 10.1.5.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.548/1.804/2.028/0.197 ms
```

✓ done

5. Check the routing table on srl1:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-instance
default route-table ipv4-unicast summary"
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-  
instance default route-table ipv4-unicast summary"
```

```
-----  
-----  
IPv4 unicast route table of network instance default  
-----  
-----
```

+-----+-----+-----+-----+-----+-----+-----+						
-+-----+-----+-----+-----+-----+-----+						
Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
		Type	Owner		Network	
	(Type)	Interfac	Next-hop	Next-hop		
					Instance	
		e	(Type)	Interfac		
				e		
+=====+=====+=====+=====+=====+=====+=====						
=+=====+=====+=====+=====+=====+						
10.1.1.0	2	local	net_inst	True	default	0
0	10.1.1.1	ethernet				
/24			_mgr			
	(direct)	-1/1.0				
10.1.1.1	2	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.1.2	2	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
10.1.2.0	3	local	net_inst	True	default	0
0	10.1.2.1	ethernet				
/24			_mgr			
	(direct)	-1/2.0				
10.1.2.1	3	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.2.2	3	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
10.1.3.0	4	local	net_inst	True	default	0
0	10.1.3.1	ethernet				

/24			_mgr			
	(direct)	-1/3.0				
10.1.3.1	4	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.3.2	4	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
10.1.4.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/2.0				
	irect/lo					
	cal)					
10.1.5.0	0	static	static_r	True	default	1
5	10.1.3.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/3.0				
	irect/lo					
	cal)					
+-----+-----+-----+-----+-----+-----+-----						
-+-----+-----+-----+-----+-----+-----+						

IPv4 routes total : 11						
IPv4 prefixes with active routes : 11						
IPv4 prefixes with active ECMP routes: 0						

Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',						
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '						
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',						
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p						
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',						
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'						
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-						
interface', 'watch', '{}']						

Deliverables

- Which pings succeeded (all should work now -- unlike lesson 02)
 - ✓ done
 - Routing table output from srl1 showing both directly connected and static routes
 - ✓ done
-

Exercise 2: Read the Routing Table

Objective: Understand routing table entries and trace a packet's path hop by hop.

✓ done

Steps

1. Examine the routing table on all 3 routers:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-instance
default route-table ipv4-unicast summary"
docker exec -it clab-routing-basics-srl2 sr_cli -c "show network-instance
default route-table ipv4-unicast summary"
docker exec -it clab-routing-basics-srl3 sr_cli -c "show network-instance
default route-table ipv4-unicast summary"
```

✓ done

```
[x]~[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]  
└─> docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-  
instance default route-table ipv4-unicast summary"
```

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
		Type	Owner		Network	
	(Type)	Interfac	Next-hop	Next-hop		
					Instance	
		e	(Type)	Interfac		
				e		

10.1.1.0	2	local	net_inst	True	default	0
0	10.1.1.1	ethernet				
/24			_mgr			
	(direct)	-1/1.0				
10.1.1.1	2	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.1.2	2	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
10.1.2.0	3	local	net_inst	True	default	0
0	10.1.2.1	ethernet				
/24			_mgr			
	(direct)	-1/2.0				
10.1.2.1	3	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.2.2	3	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					

```

| 10.1.3.0 | 4 | local | net_inst | True | default | 0
| 0 | 10.1.3.1 | ethernet | | | |
| /24 | | | _mgr | | |
| | (direct) | -1/3.0 | | | |
| 10.1.3.1 | 4 | host | net_inst | True | default | 0
| 0 | None (ex | None | | | |
| /32 | | | _mgr | | |
| | tract) | | | | |
| 10.1.3.2 | 4 | host | net_inst | True | default | 0
| 0 | None (br | | | | |
| 55/32 | | | _mgr | | |
| | oadcast) | | | | |
| 10.1.4.0 | 0 | static | static_r | True | default | 1
| 5 | 10.1.2.0 | ethernet | | | |
| /24 | | | oute_mgr | | |
| | /24 (ind | -1/2.0 | | | |
| | | | | | |
| | irect/lo | | | | |
| | | | | | |
| | cal) | | | | |
| 10.1.5.0 | 0 | static | static_r | True | default | 1
| 5 | 10.1.3.0 | ethernet | | | |
| /24 | | | oute_mgr | | |
| | /24 (ind | -1/3.0 | | | |
| | | | | | |
| | irect/lo | | | | |
| | | | | | |
| | cal) | | | | |
+-----+-----+-----+-----+-----+-----+-----+
-+-----+-----+-----+-----+-----+
-----
-----
IPv4 routes total : 11
IPv4 prefixes with active routes : 11
IPv4 prefixes with active ECMP routes: 0
-----
-----
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{}']

```

```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl2 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
```

```
-----
-----
IPv4 unicast route table of network instance default
-----
-----
```

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
	Type	Owner			Network	
	(Type)	Interfac	Next-hop	Next-hop		
					Instance	
	e	(Type)	Interfac			
				e		
+=====+=====+=====+=====+=====+=====+=====						
=+=====+=====+=====+=====+=====+						
10.1.1.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/1.0				
	irect/lo					
	cal)					
10.1.2.0	2	local	net_inst	True	default	0
0	10.1.2.2	ethernet				
/24			_mgr			
	(direct)	-1/1.0				
10.1.2.2	2	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.2.2	2	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
10.1.3.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/1.0				


```

| | | | | | |
| |irect/lo | | | |
| | | | | | |
| |cal) | | | |
| 10.1.4.0 | 3 | local | net_inst | True | default | 0
| 0 | 10.1.4.1 | ethernet | | | |
| /24 | | | _mgr | | |
| | (direct) | -1/2.0 | | |
| 10.1.4.1 | 3 | host | net_inst | True | default | 0
| 0 | None (ex | None | | | |
| /32 | | | _mgr | | |
| | tract) | | | |
| 10.1.4.2 | 3 | host | net_inst | True | default | 0
| 0 | None (br | | | | |
| 55/32 | | | _mgr | | |
| | oadcast) | | | |
| 10.1.5.0 | 0 | static | static_r | True | default | 1
| 5 | 10.1.2.0 | ethernet | | | |
| /24 | | | oute_mgr | | |
| | /24 (ind | -1/1.0 | | |
| | | | | | |
| |irect/lo | | | |
| | | | | | |
| |cal) | | | |
+-----+-----+-----+-----+-----+-----+-----+
-+-----+-----+-----+-----+-----+
-----
-----
IPv4 routes total : 9
IPv4 prefixes with active routes : 9
IPv4 prefixes with active ECMP routes: 0
-----
-----
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-
interface', 'watch', '{}']

```

```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl3 sr_cli -c "show network-
instance default route-table ipv4-unicast summary"
```

```
-----
-----
IPv4 unicast route table of network instance default
-----
-----
```

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
	Type	Owner			Network	
	(Type)	Interfac	Next-hop	Next-hop		
					Instance	
	e	(Type)	Interfac			
				e		
+=====+=====+=====+=====+=====+=====+=====						
=+=====+=====+=====+=====+=====+						
10.1.1.0	0	static	static_r	True	default	1
5	10.1.3.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/1.0				
	irect/lo					
	cal)					
10.1.2.0	0	static	static_r	True	default	1
5	10.1.3.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/1.0				
	irect/lo					
	cal)					
10.1.3.0	2	local	net_inst	True	default	0
0	10.1.3.2	ethernet				
/24			_mgr			
	(direct)	-1/1.0				
10.1.3.2	2	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					

10.1.3.2	2	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
10.1.4.0	0	static	static_r	True	default	1
5	10.1.3.0	ethernet				
/24			oute_mgr			
	/24 (ind	-1/1.0				
	irect/lo					
	cal)					
10.1.5.0	3	local	net_inst	True	default	0
0	10.1.5.1	ethernet				
/24			_mgr			
	(direct)	-1/2.0				
10.1.5.1	3	host	net_inst	True	default	0
0	None (ex	None				
/32			_mgr			
	tract)					
10.1.5.2	3	host	net_inst	True	default	0
0	None (br					
55/32			_mgr			
	oadcast)					
+-----+-----+-----+-----+-----+-----+-----+-----						
-+-----+-----+-----+-----+-----+-----+						

IPv4 routes total : 9						
IPv4 prefixes with active routes : 9						
IPv4 prefixes with active ECMP routes: 0						

Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',						
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '						
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',						
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p						
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',						
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'						
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-						
interface', 'watch', '{']						

2. For each router, identify:
- Which routes are "directly connected" (local)
 - Which routes are "static"
-  done

srl1

```
[x]~[kc-debian13-test00-Klon]~[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> docker exec -it clab-routing-basics-srl1 sr_cli
```

```
Using configuration file(s): []
```

```
Welcome to the srlinux CLI.
```

```
Type 'help' (and press <ENTER>) if you need any help using this.
```

```
--{ + running }--[ ]--
```

```
A:srl1# show network-instance default route-table ipv4-unicast summary |grep  
-i "local\|static\|prefix.*id"
```

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
10.1.1.0	2	local	net_inst	True	default	0
0	10.1.1.1	ethernet				
10.1.2.0	3	local	net_inst	True	default	0
0	10.1.2.1	ethernet				
10.1.3.0	4	local	net_inst	True	default	0
0	10.1.3.1	ethernet				
10.1.4.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
10.1.5.0	0	static	static_r	True	default	1
5	10.1.3.0	ethernet				

```
--{ + running }--[ ]--
```

```
]
```

srl2

```
[*]~[kc-debian13-test00-Klon]~[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> docker exec -it clab-routing-basics-srl2 sr_cli
```

```
Using configuration file(s): []
```

```
Welcome to the srlinux CLI.
```

```
Type 'help' (and press <ENTER>) if you need any help using this.
```

```
--{ + running }--[ ]--
```

```
A:srl2# show network-instance default route-table ipv4-unicast summary |grep
```

```
-i "local\|static\|prefix.*id"
```

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
10.1.1.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
10.1.2.0	2	local	net_inst	True	default	0
0	10.1.2.2	ethernet				
10.1.3.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
10.1.4.0	3	local	net_inst	True	default	0
0	10.1.4.1	ethernet				
10.1.5.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				

```
--{ + running }--[ ]--
```

srl3

```
[x]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl3 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl3# show network-instance default route-table ipv4-unicast summary |grep
-i "local\|static\|prefix.*id"
| Prefix | ID | Route | Route | Active | Origin | Metric |
| Pref | Next-hop | Next-hop | Backup | Backup | |
| 10.1.1.0 | 0 | static | static_r | True | default | 1 |
| 5 | 10.1.3.0 | ethernet | | | |
| 10.1.2.0 | 0 | static | static_r | True | default | 1 |
| 5 | 10.1.3.0 | ethernet | | | |
| 10.1.3.0 | 2 | local | net_inst | True | default | 0 |
| 0 | 10.1.3.2 | ethernet | | | |
| 10.1.4.0 | 0 | static | static_r | True | default | 1 |
| 5 | 10.1.3.0 | ethernet | | | |
| 10.1.5.0 | 3 | local | net_inst | True | default | 0 |
| 0 | 10.1.5.1 | ethernet | | | |
--{ + running }--[ ]--
```

3. Trace the path of a packet from host2 (10.1.4.2) to host3 (10.1.5.2):

- Step 1: host2 sends to its default gateway (10.1.4.1 = srl2)
- Step 2: srl2 looks up 10.1.5.0/24 in its routing table -- what does it find?
- Step 3: The packet arrives at srl1 -- what route does srl1 use for 10.1.5.0/24?
- Step 4: The packet arrives at srl3 -- how does it reach host3?

```
[x]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-host2 sh
/ # traceroute 10.1.5.2
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
1 10.1.4.1 (10.1.4.1) 0.963 ms 1.438 ms 1.430 ms
2 10.1.2.1 (10.1.2.1) 4.741 ms 4.817 ms 4.829 ms
3 10.1.3.2 (10.1.3.2) 5.329 ms 6.065 ms 6.073 ms
4 10.1.5.2 (10.1.5.2) 4.811 ms 4.817 ms 5.745 ms
```

4. Now trace the RETURN path (host3 -> host2):

```
[*]~[kc-debian13-test00-Klon]~[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]  
└─> docker exec -it clab-routing-basics-host3 sh  
/ # traceroute -n 10.1.4.2  
traceroute to 10.1.4.2 (10.1.4.2), 30 hops max, 60 byte packets  
1  10.1.5.1  0.906 ms  0.859 ms  0.849 ms  
2  10.1.3.1  0.696 ms  0.704 ms  0.701 ms  
3  10.1.2.2  1.753 ms  1.968 ms  2.018 ms  
4  10.1.4.2  1.167 ms  1.364 ms  1.465 ms
```

- What route does srl3 use to reach 10.1.4.0/24?

```

[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> ssh clab-routing-basics-srl3
Warning: Permanently added 'clab-routing-basics-srl3' (ED25519) to the list
of known hosts.
.....
:                               Welcome to Nokia SR Linux!                               :
:                               Open Network OS for the NetOps era.                               :
:                               :                               :
:   This is a freely distributed official container image.   :
:                               Use it - Share it                               :
:                               :                               :
: Get started: https://learn.srlinux.dev                               :
: Container:   https://go.srlinux.dev/container-image                               :
: Docs:       https://doc.srlinux.dev/24-10                               :
: Rel. notes: https://doc.srlinux.dev/rn24-10-1                               :
: YANG:       https://yang.srlinux.dev/v24.10.1                               :
: Discord:    https://go.srlinux.dev/discord                               :
: Contact:    https://go.srlinux.dev/contact-sales                               :
:                               :                               :
.....
[ ]
Last login: Fri May 22 03:46:17 2026 from 3fff:172:20:20::1
Using configuration file(s): ['/home/admin/.srlinuxrc']
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
[ ]
--{ + running }--[ ]--
A:srl3# traceroute -n 10.2.4.2
Using network instance mgmt
traceroute to 10.2.4.2 (10.2.4.2), 30 hops max, 60 byte packets
 1  172.20.20.1  8.029 ms  7.505 ms  7.496 ms
 2  192.168.178.250  8.461 ms  8.464 ms  8.460 ms
 3  192.168.4.1  10.447 ms  12.483 ms  10.428 ms
 4  * * *
[.]
 8  * * *
^CCommand execution aborted : 'traceroute -n 10.2.4.2'
[ ]
--{ + running }--[ ]--

```

Something strange happens here. Because the router `srl3` does correctly route packages from `host3` to `host2` ?!


```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> ssh clab-routing-basics-srl3
```

```
Warning: Permanently added 'clab-routing-basics-srl3' (ED25519) to the list
of known hosts.
```

.....		
:	Welcome to Nokia SR Linux!	:
:	Open Network OS for the NetOps era.	:
:		:
:	This is a freely distributed official container image.	:
:	Use it - Share it	:
:		:
:	Get started: https://learn.srlinux.dev	:
:	Container: https://go.srlinux.dev/container-image	:
:	Docs: https://doc.srlinux.dev/24-10	:
:	Rel. notes: https://doc.srlinux.dev/rn24-10-1	:
:	YANG: https://yang.srlinux.dev/v24.10.1	:
:	Discord: https://go.srlinux.dev/discord	:
:	Contact: https://go.srlinux.dev/contact-sales	:
.....		

```
Last login: Fri May 22 03:49:27 2026 from 3fff:172:20:20::1
```

```
Using configuration file(s): ['/home/admin/.srlinuxrc']
```

```
Welcome to the srlinux CLI.
```

```
Type 'help' (and press <ENTER>) if you need any help using this.
```

```
--{ + running }--[ ]--
```

```
A:srl3# show network-instance mgmt route-table ipv4-unicast summary
```

```
IPv4 unicast route table of network instance mgmt
```

+-----+-----+-----+-----+-----+-----+
 -+-----+-----+-----+-----+-----+-----+

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
		Type	Owner		Network	
	(Type)	Interfac	Next-hop	Next-hop		
					Instance	
		e	(Type)	Interfac		
				e		

[illegible]

0.0.0.0/	1	dhcp	dhcp_cli	True	mgmt	0
5	172.20.2	mgmt0.0				

0			ent_mgr			
	0.1					
	(direct)					
172.20.2	0	linux	linux_mg	False	mgmt	0
5	172.20.2	mgmt0.0				
0.0/24			r			
	0.0					
	(direct)					
172.20.2	1	local	net_inst	True	mgmt	0
0	172.20.2	mgmt0.0				
0.0/24			_mgr			
	0.13					
	(direct)					
172.20.2	1	host	net_inst	True	mgmt	0
0	None (ex	None				
0.13/32			_mgr			
	tract)					
172.20.2	1	host	net_inst	True	mgmt	0
0	None (br					
0.255/32			_mgr			
	oadcast)					
+-----+-----+-----+-----+-----+-----+-----						
-+-----+-----+-----+-----+-----+						

IPv4 routes total : 5						
IPv4 prefixes with active routes : 4						
IPv4 prefixes with active ECMP routes: 0						

[						
--{ + running }--[]--						

The problem seems to be, that the connected `host3` does use an other Routing tabel (`default`) than the router self emitted packages via `ping -c 3 10.1.4.2`. These packages seem to use the routing table calles `mgmt`.

Deliverables

- Routing table screenshots from all 3 routers
- ✓ done

- Written hop-by-hop trace for host2 -> host3 AND the return path

✓ done

Exercise 3: Break/Fix -- Missing Route

Objective: Understand what happens when a router is missing a route and why both directions can break even when only one router is affected.

Setup (break it)

```
docker exec -it clab-routing-basics-srl2 sr_cli
```

Inside SR Linux:

```
enter candidate
delete / network-instance default static-routes route 10.1.5.0/24
delete / network-instance default next-hop-groups group nhg-10-1-5-0-24
commit now
exit
```

```
└─> docker exec -it clab-routing-basics-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl2# enter candidate
delete / network-instance default static-routes route 10.1.5.0/24
delete / network-instance default next-hop-groups group nhg-10-1-5-0-24
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[ ]--
A:srl2#
EOF encountered
```

Symptom

```
# This FAILS -- host2 cannot reach host3
docker exec clab-routing-basics-host2 ping -c 3 -W 5 10.1.5.2

# This ALSO FAILS -- host3 cannot reach host2 (why?)
docker exec clab-routing-basics-host3 ping -c 3 -W 5 10.1.4.2

# But this WORKS -- host1 can still reach host3
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host2 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.4.1 icmp_seq=1 Destination Net Unreachable
From 10.1.4.1 icmp_seq=2 Destination Net Unreachable
From 10.1.4.1 icmp_seq=3 Destination Net Unreachable

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2032ms

[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host3 ping -c 3 -W 5 10.1.4.2
PING 10.1.4.2 (10.1.4.2) 56(84) bytes of data.

--- 10.1.4.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2044ms

[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.590 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.353 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.258 ms

--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2037ms
rtt min/avg/max/mdev = 0.258/0.400/0.590/0.139 ms
```

Your Task

1. Check srl2's routing table -- what's missing?

```
docker exec -it clab-routing-basics-srl2 sr_cli -c "show network-instance  
default route-table ipv4-unicast summary"
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> ssh clab-routing-basics-srl2
```

```
Warning: Permanently added 'clab-routing-basics-srl2' (ED25519) to the list
of known hosts.
```

.....	:		:
:		Welcome to Nokia SR Linux!	:
:		Open Network OS for the NetOps era.	:
:			:
:		This is a freely distributed official container image.	:
:		Use it - Share it	:
:			:
:	Get started:	https://learn.srlinux.dev	:
:	Container:	https://go.srlinux.dev/container-image	:
:	Docs:	https://doc.srlinux.dev/24-10	:
:	Rel. notes:	https://doc.srlinux.dev/rn24-10-1	:
:	YANG:	https://yang.srlinux.dev/v24.10.1	:
:	Discord:	https://go.srlinux.dev/discord	:
:	Contact:	https://go.srlinux.dev/contact-sales	:
.....	:		:

```
Using configuration file(s): ['/home/admin/.srlinuxrc']
```

```
Welcome to the srlinux CLI.
```

```
Type 'help' (and press <ENTER>) if you need any help using this.
```

```
--{ + running }--[ ]--
```

```
A:srl2# show network-instance default route-table ipv4-unicast summary
```

```
IPv4 unicast route table of network instance default
```

Prefix	ID	Route	Route	Active	Origin	Metric
Pref	Next-hop	Next-hop	Backup	Backup		
		Type	Owner		Network	
	(Type)	Interfac	Next-hop	Next-hop		
					Instance	
		e	(Type)	Interfac		
				e		
10.1.1.0	0	static	static_r	True	default	1
5	10.1.2.0	ethernet				
/24			oute mgr			

2. Explain why both host2->host3 AND host3->host2 fail, even though only srl2 is missing a route (hint: trace both the forward path and the return path for each ping).

The router `srl2` is missing the default route to reach `10.1.5.2`

```
[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]  
└─> docker exec -it clab-routing-basics-host2 sh  
/ # traceroute -n 10.1.5.2  
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets  
1  10.1.4.1  0.555 ms !N  0.531 ms !N  0.489 ms !N
```

The router `srl3` has intact routes, but the answers from `srl2` never come back to `host3` (via `srl3`) because they never are sent back to `srl3`

```
[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]  
└─> docker exec -it clab-routing-basics-host3 sh  
/ # traceroute -n 10.1.4.2  
traceroute to 10.1.4.2 (10.1.4.2), 30 hops max, 60 byte packets  
1  10.1.5.1  1.485 ms  1.453 ms  1.447 ms  
2  10.1.3.1  1.331 ms  1.344 ms  1.341 ms  
3  * * *  
[..]  
7  * * *
```

3. Fix by re-adding the route:

```
docker exec -it clab-routing-basics-srl2 sr_cli
```

Inside SR Linux:

```
enter candidate  
set / network-instance default next-hop-groups group nhg-10-1-5-0-24  
admin-state enable  
set / network-instance default next-hop-groups group nhg-10-1-5-0-24  
nexthop 1 ip-address 10.1.2.1  
set / network-instance default static-routes route 10.1.5.0/24 admin-  
state enable  
set / network-instance default static-routes route 10.1.5.0/24 next-hop-  
group nhg-10-1-5-0-24  
commit now  
exit
```



```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl2 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl2# enter candidate
  set / network-instance default next-hop-groups group nhg-10-1-5-0-24
admin-state enable
  set / network-instance default next-hop-groups group nhg-10-1-5-0-24
nexthop 1 ip-address 10.1.2.1
  set / network-instance default static-routes route 10.1.5.0/24 admin-state
enable
  set / network-instance default static-routes route 10.1.5.0/24 next-hop-
group nhg-10-1-5-0-24
  commit now
  exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[ ]--
A:srl2#
EOF encountered
```

4. Verify the fix:

```
docker exec clab-routing-basics-host2 ping -c 3 10.1.5.2
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host2 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=61 time=0.800 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=61 time=0.767 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=61 time=0.637 ms
[ ]
--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2049ms
rtt min/avg/max/mdev = 0.637/0.734/0.800/0.070 ms
```

Deliverables

- Explanation of why the missing route breaks both directions (forward path vs return path)
-  done

- The diagnostic commands you used

✓ done

Exercise 4: Break/Fix -- Wrong Next-Hop (Black Hole)

Objective: Understand what happens when a static route points to a non-existent next-hop.

Setup (break it)

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.3.99
commit now
exit
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl1# enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.3.99
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[ ]--
```

Symptom

```
# host1 cannot reach host3
docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
```

```
[x]~[kc-debian13-test00-Klon]~[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]  
└─> docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2  
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.  
  
--- 10.1.5.2 ping statistics ---  
3 packets transmitted, 0 received, 100% packet loss, time 2033ms
```

Your Task

1. Check srl1's routing table -- does the route exist?

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-instance  
default route-table ipv4-unicast summary"
```

```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
↳ docker exec -it clab-routing-basics-srl1 sr_cli -c "show network-  
instance default route-table ipv4-unicast summary"
```

```
-----  
-----
```

```
IPv4 unicast route table of network instance default
```

```
-----  
-----
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
-+-----+-----+-----+-----+-----+-----+
```

Prefix	ID	Route	Route	Active	Origin	Metric
--------	----	-------	-------	--------	--------	--------

Pref	Next-hop	Next-hop	Backup	Backup	
------	----------	----------	--------	--------	--

		Type	Owner		Network	
--	--	------	-------	--	---------	--

	(Type)	Interfac	Next-hop	Next-hop	
--	--------	----------	----------	----------	--

					Instance	
--	--	--	--	--	----------	--

		e	(Type)	Interfac	
--	--	---	--------	----------	--

|--|--|--|--|--|--|--|

				e	
--	--	--	--	---	--

```
+=====+=====+=====+=====+=====+=====+=====+=====
```

```
=+=====+=====+=====+=====+=====+
```

10.1.1.0	2	local	net_inst	True	default	0
----------	---	-------	----------	------	---------	---

0	10.1.1.1	ethernet			
---	----------	----------	--	--	--

/24			_mgr			
-----	--	--	------	--	--	--

	(direct)	-1/1.0			
--	----------	--------	--	--	--

10.1.1.1	2	host	net_inst	True	default	0
----------	---	------	----------	------	---------	---

0	None (ex	None			
---	----------	------	--	--	--

/32			_mgr			
-----	--	--	------	--	--	--

	tract)				
--	--------	--	--	--	--

10.1.1.2	2	host	net_inst	True	default	0
----------	---	------	----------	------	---------	---

0	None (br				
---	----------	--	--	--	--

55/32			_mgr			
-------	--	--	------	--	--	--

	oadcast)				
--	----------	--	--	--	--

10.1.2.0	3	local	net_inst	True	default	0
----------	---	-------	----------	------	---------	---

0	10.1.2.1	ethernet			
---	----------	----------	--	--	--

/24			_mgr			
-----	--	--	------	--	--	--

	(direct)	-1/2.0			
--	----------	--------	--	--	--

10.1.2.1	3	host	net_inst	True	default	0
----------	---	------	----------	------	---------	---

0	None (ex	None			
---	----------	------	--	--	--

/32			_mgr			
-----	--	--	------	--	--	--

	tract)				
--	--------	--	--	--	--

10.1.2.2	3	host	net_inst	True	default	0
----------	---	------	----------	------	---------	---

0	None (br				
---	----------	--	--	--	--

55/32			_mgr			
-------	--	--	------	--	--	--

	oadcast)				
--	----------	--	--	--	--

10.1.3.0	4	local	net_inst	True	default	0
----------	---	-------	----------	------	---------	---

0	10.1.3.1	ethernet			
---	----------	----------	--	--	--

Problem: My environment does not reach the broken state that is described in the solution

```
~{ + running }~{ ]--
A:srl1# show network-instance default route-table ipv4-unicast summary
IPv4 unicast route table of network instance default
```

Prefix	ID	Route Type	Route Owner	Active	Origin Network Instance	Metric	Pref	Next-hop (Type)	Next-hop Interface
10.1.1.0/24	2	local	net_inst_mgr	True	default	0	0	10.1.1.1 (direct)	ethernet-1/1.0
10.1.1.1/32	2	host	net_inst_mgr	True	default	0	0	None (extract)	None
10.1.1.255/32	2	host	net_inst_mgr	True	default	0	0	None (broadcast)	None
10.1.2.0/24	3	local	net_inst_mgr	True	default	0	0	10.1.2.1 (direct)	ethernet-1/2.0
10.1.2.1/32	3	host	net_inst_mgr	True	default	0	0	None (extract)	None
10.1.2.255/32	3	host	net_inst_mgr	True	default	0	0	None (broadcast)	None
10.1.3.0/24	4	local	net_inst_mgr	True	default	0	0	10.1.3.1 (direct)	ethernet-1/3.0
10.1.3.1/32	4	host	net_inst_mgr	True	default	0	0	None (extract)	None
10.1.3.255/32	4	host	net_inst_mgr	True	default	0	0	None (broadcast)	None
10.1.4.0/24	0	static	static_route_mgr	True	default	1	5	10.1.2.0/24 (indirect/local)	ethernet-1/2.0
10.1.5.0/24	0	static	static_route_mgr	True	default	1	5	10.1.3.0/24 (indirect/local)	ethernet-1/3.0

```
IPv4 routes total : 11
IPv4 prefixes with active routes : 11
IPv4 prefixes with active ECMP routes: 0
```

I never get to the point where the next-hop is literally **10.1.3.99** evenb if I executed all commands as stated above

```
**srl1's routing table after the change:**
~{
A:srl1# show network-instance default route-table ipv4-unicast summary
+-----+-----+-----+-----+-----+
| Prefix | Type | Route-Owner | Next-hop | Metric |
+=====+=====+=====+=====+=====+
| 10.1.1.0/24 | local | net_inst | ethernet-1/1.0 | 0 |
| 10.1.2.0/24 | local | net_inst | ethernet-1/2.0 | 0 |
| 10.1.3.0/24 | local | net_inst | ethernet-1/3.0 | 0 |
| 10.1.4.0/24 | static | static | 10.1.2.2 | 1 |
| 10.1.5.0/24 | static | static | 10.1.3.99 | 1 |
+-----+-----+-----+-----+-----+
~{
```

2. The route exists but points to 10.1.3.99. Check the ARP table:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show arpnd arp-entries"
```

```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> docker exec -it clab-routing-basics-srl1 sr_cli -c "show arpnd arp-  
entries"
```

```
+-----+-----+-----+-----+-----+-----+  
-----+-----+  
| Interface | Subinterfac | Neighbor | Origin | Link layer |  
address | Expiry |  
| | e | | | |  
| | |  
+=====+=====+=====+=====+=====+=====+  
=====+=====+  
| ethernet- | 0 | 10.1.1.2 | dynamic |  
AA:C1:AB:C5:FA:06 | 3 hours from now |  
| 1/1 | | | |  
| | |  
| ethernet- | 0 | 10.1.2.2 | dynamic |  
1A:33:04:FF:00:01 | 2 hours from now |  
| 1/2 | | | |  
| | |  
| mgmt0 | 0 | 172.20.20.1 | dynamic |  
BA:4B:B5:4E:81:AA | 2 hours from now |  
+-----+-----+-----+-----+-----+-----+  
-----+-----+  
-----+  
-----+  
Total entries : 3 (0 static, 3 dynamic)  
-----+  
-----+  
Parsing error: Unknown token 'commit'. Options are ['!', '#', '/', 'acl',  
'back', 'bash', 'bfd', 'cls', 'date', 'diff', 'echo', 'enter', '  
environment', 'exit', 'file', 'help', 'history', 'info', 'interface',  
'list', 'monitor', 'network-instance', 'passwd', 'ping', 'ping6', 'p  
latform', 'pwc', 'qos', 'quit', 'references', 'routing-policy', 'save',  
'show', 'source', 'ssh', 'system', 'tcptraceroute', 'tech-support'  
, 'tools', 'traceroute', 'traceroute6', 'tree', 'tunnel', 'tunnel-  
interface', 'watch', '{}']
```

Problem: My environment does not reach the broken state that is described in the solution

```
**ARP table on srl1:**
```

```
\\
```

```
A:srl1# show arpnd arp-entries
```

Interface	IP Addr	MAC	State
e1-1.0	10.1.1.2	aa:c1:...	reachable
e1-2.0	10.1.2.2	aa:c1:...	reachable
e1-3.0	10.1.3.2	aa:c1:...	reachable
e1-3.0	10.1.3.99		incomplete

3. Explain why a valid-looking route can still black-hole traffic.

- Because a valid looking route can still have an invalid target like `10.1.3.99`

4. Fix by restoring the correct next-hop:

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
```

```
set / network-instance default next-hop-groups group nhg-10-1-5-0-24
```

```
nexthop 1 ip-address 10.1.3.2
```

```
commit now
```

```
exit
```

5. Verify:

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
```


Deliverables

- Explanation of black hole routing
 - If the route a router would need to send packages bag does not exist, and the returning packages go throu the default route (and never return to the originating host)
 - ARP output showing failed resolution for 10.1.3.99
 - I cant complete this excersise, because the setup that I should have for my experimentation does not show up after following all instructions
-

Exercise 5: Break/Fix -- Routing Loop

Objective: Observe what happens when two routers send traffic back and forth in a loop, and understand how TTL prevents infinite loops.

Setup (break it)

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.2.2
commit now
exit
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl1# enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24 nexthop
1 ip-address 10.1.2.2
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[ ]--
A:srl1#
```

Now srl1 sends 10.1.5.0/24 traffic to srl2, and srl2 sends it back to srl1 (via its existing route through the hub).

Symptom

```
# Ping fails
docker exec clab-routing-basics-host1 ping -c 3 -W 10 10.1.5.2
[ ]
# Traceroute reveals the loop
docker exec clab-routing-basics-host1 traceroute -n -w 2 10.1.5.2
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> docker exec clab-routing-basics-host1 ping -c 3 -W 10 10.1.5.2
```

```
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
```

```
From 10.1.2.1 icmp_seq=1 Redirect Network(New nexthop: 10.1.2.2)
```

```
From 10.1.2.1 icmp_seq=1 Redirect Network(New nexthop: 10.1.2.2)
```

```
From 10.1.2.1 icmp_seq=1 Redirect Network(New nexthop: 10.1.2.2)
```

```
--- 10.1.5.2 ping statistics ---
```

```
1 packets transmitted, 0 received, +3 errors, 100% packet loss, time 0ms
```

```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]
```

```
└─> docker exec clab-routing-basics-host1 traceroute -n -w 2 10.1.5.2
```

```
traceroute to 10.1.5.2 (10.1.5.2), 30 hops max, 60 byte packets
```

```
1  10.1.1.1  1.353 ms  1.336 ms  1.331 ms
```

```
2  10.1.2.2  1.416 ms  1.416 ms  1.415 ms
```

```
3  10.1.2.1  2.414 ms  2.420 ms  2.417 ms
```

```
4  10.1.2.2  1.818 ms  1.911 ms  2.423 ms
```

```
5  * * *
```

```
6  10.1.2.2  2.533 ms  2.077 ms  2.076 ms
```

```
7  * * *
```

```
8  10.1.2.2  2.687 ms  * *
```

```
9  * * *
```

```
10 * * *
```

```
11 * * *
```

```
12 10.1.2.2 12.756 ms 12.767 ms 12.766 ms
```

```
13 * * *
```

```
14 10.1.2.2 12.735 ms 11.977 ms 11.970 ms
```

```
15 * * *
```

```
16 10.1.2.2 10.881 ms 10.781 ms 10.626 ms
```

```
17 * * *
```

```
18 10.1.2.2 1.858 ms * *
```

```
19 * * *
```

```
20 * 10.1.2.2 4.004 ms 3.850 ms
```

```
21 * * *
```

```
22 10.1.2.2 4.708 ms 4.709 ms 1.732 ms
```

```
23 * * *
```

```
24 10.1.2.2 1.898 ms 0.968 ms 0.903 ms
```

```
25 * * *
```

```
26 10.1.2.2 13.006 ms 13.024 ms *
```

```
27 * * *
```

```
28 * * *^C
```

Your Task

1. Run traceroute and observe the alternating hops between srl1 and srl2.
2. Explain: What is TTL? Why does the packet eventually stop?
 - The `TTL` is the number of maximum hops a package may take, before it is discarded
3. Fix by restoring srl1's correct next-hop:

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24
nexthop 1 ip-address 10.1.3.2
commit now
exit
```

```
[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl1# enter candidate
set / network-instance default next-hop-groups group nhg-10-1-5-0-24
nexthop 1 ip-address 10.1.3.2
commit now
exit
All changes have been committed. Leaving candidate mode.
[]
```

4. Verify:

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
```

```
[x]~[kc-debian13-test00-Klon]~[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics/ansible]  
└─> docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2  
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.  
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.786 ms  
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.478 ms  
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.382 ms  
  
--- 10.1.5.2 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2038ms  
rtt min/avg/max/mdev = 0.382/0.548/0.786/0.172 ms
```

Deliverables

- Traceroute output showing the routing loop pattern
✓ done
- Explanation of TTL and how it prevents infinite loops
✓ done

Exercise 6: Break/Fix -- Unreachable Next-Hop (Link Down)

Objective: Understand that a route can exist in the routing table even when the physical path is broken.

Setup (break it)

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate  
set / interface ethernet-1/3 admin-state disable  
commit now  
exit
```

Symptom

```
# host1 cannot reach host3
docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
```

```
└─> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl1# enter candidate
set / interface ethernet-1/3 admin-state disable
commit now
exit
All changes have been committed. Leaving candidate mode.
--{ + running }--[ ]--
A:srl1#
EOF encountered
[*]--[kc-debian13-test00-Klon]--[kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/ansible]
└─> docker exec clab-routing-basics-host1 ping -c 3 -W 5 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
From 10.1.1.1 icmp_seq=1 Destination Net Unreachable
From 10.1.1.1 icmp_seq=2 Destination Net Unreachable
^C
```

Your Task

1. Check interface status on srl1:

```
docker exec -it clab-routing-basics-srl1 sr_cli -c "show interface brief"
```

```
A:srl1# show interface brief
```

+-----+-----+-----+-----+			
+-----+-----+-----+-----+			
Port	Admin State	Oper State	Speed
Type	Description		
+=====+=====+=====+=====+			
+=====+=====+=====+=====+			
ethernet-1/1	enable	up	25G
ethernet-1/2	enable	up	25G
ethernet-1/3	disable	down	25G
ethernet-1/4	disable	down	25G

2. Notice ethernet-1/3 is admin-disabled. The route to 10.1.5.0/24 may still exist but the next-hop is unreachable.
3. Explain: Why can a route exist even when the link is down?
 - (From solutions:) *Static routes are manually configured entries. The router does not automatically remove them when a link goes down. Dynamic routing protocols like BGP or OSPF would detect the link failure and withdraw the route from the table, but static routes are "dumb" -- they stay in the table regardless of link state.*
4. Fix by re-enabling the interface:

```
docker exec -it clab-routing-basics-srl1 sr_cli
```

Inside SR Linux:

```
enter candidate
set / interface ethernet-1/3 admin-state enable
commit now
exit
```

5. Verify:

```
docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
```

```
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/an
sible]
└─> docker exec -it clab-routing-basics-srl1 sr_cli
Using configuration file(s): []
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ + running }--[ ]--
A:srl1# enter candidate
  set / interface ethernet-1/3 admin-state enable
  commit now
  exit
  [ ]
All changes have been committed. Leaving candidate mode.
--{ + running }--[ ]--
A:srl1#
EOF encountered
[ ]
[*]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-
kubecraft/lessons/clab/03-routing-basics/an
sible]
└─> docker exec clab-routing-basics-host1 ping -c 3 10.1.5.2
PING 10.1.5.2 (10.1.5.2) 56(84) bytes of data.
64 bytes from 10.1.5.2: icmp_seq=1 ttl=62 time=0.481 ms
64 bytes from 10.1.5.2: icmp_seq=2 ttl=62 time=0.260 ms
64 bytes from 10.1.5.2: icmp_seq=3 ttl=62 time=0.248 ms
[ ]
--- 10.1.5.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2009ms
rtt min/avg/max/mdev = 0.248/0.329/0.481/0.107 ms
```

Deliverables

- Interface status output showing the disabled link
 done
- Explanation of why route existence does not guarantee path availability
 done

Cleanup

After completing all exercises:


```
# Destroy the lab
containerlab destroy -t topology/lab.clab.yml --cleanup
# Verify
docker ps | grep clab
```

Validation

Run the automated tests:

```
pytest tests/ -v
```

```
[x]-[kc-debian13-test00-Klon]-[/kubecraft/github_rbeckesch_DrewElliot-  
kubecraft/lessons/clab/03-routing-basics]
```

```
└─> pytest tests/ -v
```

```
===== test session starts
```

```
=====
```

```
platform linux -- Python 3.13.5, pytest-8.3.5, pluggy-1.5.0 --
```

```
/usr/bin/python3
```

```
cachedir: .pytest_cache
```

```
rootdir: /kubecraft/github_rbeckesch_DrewElliot-kubecraft
```

```
configfile: pyproject.toml
```

```
plugins: anyio-4.8.0
```

```
collected 22 items
```

```
tests/test_routing_basics.py::TestEnvironment::test_containerlab_installed
```

```
PASSED [ 4%]
```

```
tests/test_routing_basics.py::TestEnvironment::test_docker_available PASSED
```

```
[ 9%]
```

```
tests/test_routing_basics.py::TestEnvironment::test_ansible_installed PASSED
```

```
[ 13%]
```

```
tests/test_routing_basics.py::TestEnvironment::test_topology_file_exists
```

```
PASSED [ 18%]
```

```
tests/test_routing_basics.py::TestTopologyStructure::test_has_correct_name
```

```
PASSED [ 22%]
```

```
tests/test_routing_basics.py::TestTopologyStructure::test_has_six_nodes
```

```
PASSED [ 27%]
```

```
tests/test_routing_basics.py::TestTopologyStructure::test_has_five_links
```

```
PASSED [ 31%]
```

```
tests/test_routing_basics.py::TestTopologyStructure::test_no_latest_tags
```

```
PASSED [ 36%]
```

```
tests/test_routing_basics.py::TestTopologyStructure::test_routers_have_mgmt_
```

```
ips PASSED [ 40%]
```

```
tests/test_routing_basics.py::TestTopologyStructure::test_hosts_have_exec
```

```
PASSED [ 45%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_inventory_exists
```

```
PASSED [ 50%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_playbook_exists
```

```
PASSED [ 54%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_interface_template_
```

```
exists PASSED [ 59%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_routes_template_exi
```

```
sts PASSED [ 63%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_host_vars_exist
```

```
PASSED [ 68%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_host_vars_have_rout
```

```
es PASSED [ 72%]
```

```
tests/test_routing_basics.py::TestAnsibleStructure::test_inventory_has_three
```

```
_routers PASSED [ 77%]
```

```
tests/test_routing_basics.py::TestLabDeployment::test_lab_deploys PASSED
[ 81%]
tests/test_routing_basics.py::TestLabDeployment::test_six_containers_running
PASSED [ 86%]
tests/test_routing_basics.py::TestConnectivity::test_host1_to_host2 PASSED
[ 90%]
tests/test_routing_basics.py::TestConnectivity::test_host1_to_host3 PASSED
[ 95%]
tests/test_routing_basics.py::TestConnectivity::test_host2_to_host3 PASSED
[100%]

```

```
===== 22 passed in 306.88s (0:05:06)
=====
```

Completion Checklist

- ✓ Exercise 1: Deployed lab, applied Ansible config, verified end-to-end connectivity
- ✓ Exercise 2: Read routing tables, traced packet path hop by hop
- ✓ Exercise 3: Diagnosed and fixed missing route (return path analysis)
- ✓ Exercise 4: Diagnosed and fixed wrong next-hop (black hole)
- ✓ Exercise 5: Diagnosed and fixed routing loop (TTL)
- ✓ Exercise 6: Diagnosed and fixed unreachable next-hop (link down)

Next Steps

Commit your exercises to your fork and proceed to Lesson 4 (coming soon).